SARATH PRABHAVU J.

Tolouse, France | Kerala, India | sarathprabhav@gmail.com | +33-745240872

Objective

Seeking admission to a PhD programme in Space Science to deepen my expertise, contribute to impactful research, and integrate my passion for public outreach to inspire future generation.

Research Experience

Understanding the space weather effects on exospheric escape from Earth. — Remote, Aug 2025 – Ongoing

- **Guided by:** Dr. Dolon Bhattacharya,Research Scientist, LASP
- Study of Upper atmosphere using Lyman Alpha Emission
- Exploring solar activity effects on the upper atmospheres of terrestrial planets
- Explores the changes incurred by the exosphere of Earth when a strong solar storm impacts the planet, resulting in a large geomagnetic storm event with a highly negative Dst index
- Using the NRLMSIS model to simulate upper-atmospheric parameters and validating model outputs with GOLD satellite measurements..

High Energy Astrophysics- Gamma Ray Bursts — Kerala, India, Oct 2024 – Feb 2025

- **Guided by:** Dr Resmi L., Associate Professor, Indian Institute of Space Science and Technology.
- Studied how GRB hardness and duration vary with the observer's viewing angle.
- Modelled single-pulse GRBs using a geometrically and optically thin top-hat jet model.
- Simulated bursts with isotropic viewing angles and computed pulse durations and spectral hardness in FERMI-GBM bands.
- Established a strong dependence of spectral hardness on viewing angle (for broken power-law spectra), with a less significant impact on duration.
- Published this work in The Astrophysical Journal (March 2025).

International Liquid Mirror Telescope (ILMT), ARIES – Project Associate — Nainital, India, Apr 2024 – Sept 2024

- Gained hands-on experience in observational astronomy through ILMT operations.
- Strengthened knowledge in telescope instrumentation and troubleshooting.
- Improved ability to work effectively in an international collaboration
- Developed a web app to display ILMT FITS header data.

Education

M.Phil. Cochin University of Science and Technology, Kerala, India — Jan 2021 – Dec 2022

- GPA: 3.6/4.00
- **Electives:** Astrophysics, Cosmology
- **Thesis:** Estimation of Magnetic Field in GRB Afterglows

M.Sc University of Kerala, Kerala, India — Aug 2015

- Grade: 74.4%
- **Courses:** Classical Mechanics, Quantum Mechanics, Statistical Mechanics, Mathematical Physics
- **Project:** Estimation of Strömgren Sphere Radius around Massive Stars

B.Sc **University of Kerala**, Kerala, India — May 2012

- Grade: 87%

Publications

Effect of Viewing Angle in Gamma-Ray Burst Properties — Mar 2025

Sreelakshmi P. Chakyar, ***Sarath Prabhavu J.***, Lekshmi Resmi
The Astrophysical Journal, Volume 982, Number 1
10.3847/1538-4357/adb1e0

The 4 m International Liquid Mirror Telescope: Construction, Operation, and Science — Feb 2025

J. Surdej, P. Hickson, K. Misra, D. Banerjee, ***S. Prabhavu***, K. Pranshu, M. Kharayat, et al.
A&A Volume 694, Article Number: A80
10.1051/0004-6361/202452667

Projects & Thesis

Constraining the Magnetic-Field Energy Fraction in Gamma-Ray Burst Afterglows via Parameter Estimation — Dec 2022

Supervisor: Dr Resmi L., *Indian Institute of Space Science and Technology*

- Afterglow theory converted to a computational model
- Used Bayesian Parameter Estimation techniques.
- Multi-wavelength data employed for parameter fitting.

Radial Velocity Method of Exoplanet Detection – 51 Pegasi b — Apr 2019

Guided by: Prof. Anand Narayan, Indian Institute of Space Science and Technology

- Applied the radial velocity technique using observational datasets to replicate detection of exoplanet 51 Pegasi b.

Estimation of Strömgren Sphere Radius Around Massive Stars — Aug 2015

Guided by: Prof. Anand Narayan, *Indian Institute of Space Science and Technology*

- Calculated size of ionized H II regions around massive stars.

Teaching & Outreach Experience

Assistant Professor in Physics (Contract) Department of Physics, SN College Varkala — Kerala, India, Sept 2017 – Mar 2018 & Jul – Nov 2019

- Taught undergraduate theory and practical physics courses.
- Subjects: Vector Calculus, Thermodynamics, C++ Programming, Electricity and Magnetism.

Assistant Professor in Physics (Contract) Department of Physics, KSMDB College, Sasthamcotta — Kerala, India, Jul 2016 – Mar 2017

- Delivered undergraduate lectures in Mechanics, Thermodynamics, Electricity & Magnetism, and Introductory Quantum Physics and handled lab sessions

Astronomy Edcuator & Sceinece Communicator — International, 2014 – Present

- Astronomy Resource Person in India and the Middle East.
- Conducted sunspot observation & stargazing sessions for school students and Public.
- Host of an Astronomy TV show on the Kerala Government's TV channel
- Skilled at communicating science to audiences ranging from children to adults.
- Served as resource person and state-level coordinator for the Amateur Astronomers Organisation.

Skills

Programming Languages

- **Python** (primary language):
  - **Major Packages used:** Pymsis, NetCDF, PyHDF, Astropy, Flask, Ultranest, Matplotlib, Pandas, BeautifulSoup
  - Satellite Data Analysis: CloudSAT, COSMIC-1
  - GRB Afterglow modelling & Bayesian Parameter Estimation
  - Flask based Web development
- **Matlab:** Used for satellite data analysis.
- **C++, C:** Applied in academic mini-projects.
- **Adobe Photoshop, Illustrator, Indesign, Microsoft Office, LaTeX**

Database Management

- **PostgreSQL/SQL:** Used to store FITS header data for ILMT project.
  Integrated with Flask-based web applications for dynamic data retrieval.

Astrophotography

- Intermediate deep-sky astrophotographer with 2+ years of experience
- Second runner-up in ISRO's National Astrophotography Contest
- Skilled in post-processing of astrophotographs

Public Speaking & Science Communication

- Delivered over 100 public stargazing sessions across India & UAE
- Organised and led workshops, lectures, and outreach events for students and educators
- Authored popular science articles in regional language for both print and online publications